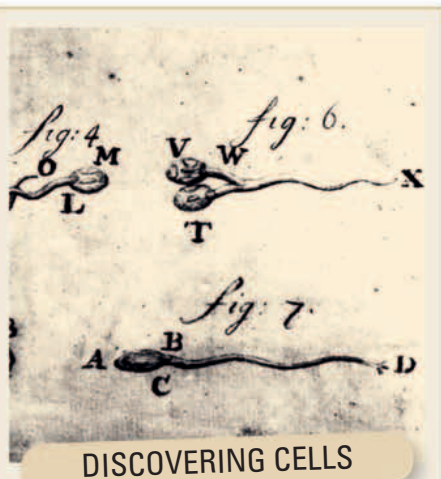


Inside cells

Right now, the trillions of cells inside your body are buzzing with activity. Inside each tiny cell there are many parts that make substances, release energy, and recycle worn-out materials. Although body cells come in many shapes and sizes, they all share the same basic structure.

IN CHARGE

Except for red blood cells, every cell in your body has a control center called the nucleus. It contains the essential instructions to build, maintain, and run the cell. Outside the controlling nucleus is the cytoplasm, a clear, jellylike liquid that contains cell parts called organelles (“tiny organs”). These organelles have their own specific task.



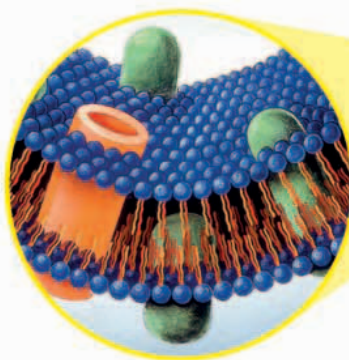
DISCOVERING CELLS

In the 1670s, Dutch textile merchant Antonie van Leeuwenhoek made some amazing discoveries using his simple but powerful microscope. His interest in studying living things led to the discovery of bacteria, red blood cells, and sperm (shown here in his drawing).

Ribosomes are tiny structures that make important substances called proteins, using instructions received from the nucleus.

The nucleus is the largest organelle within the cell.

Lysosomes break down and recycle worn-out organelles.



▲ **CELL MEMBRANE** *Guarding the contents of the cell is a protective layer called the cell membrane. It consists of various parts that help control the movement of substances into and out of the cell.*

▲ **CLOSE-UP VIEW** *This model of a typical cell shows the features that can be seen using a powerful electron microscope.*